EVALUATING CRITICAL SUCCESS FACTORS ACCORDING TO INTEGRATION MATURITY LEVEL

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ABSTRACT

The integration projects are research topics for their complexity at the present time. Their rates of success are not very encouraging, joined at their high costs. For it, *in this work a evaluation of a set of Critical Success Factors (CSFs) are proposed, according to the maturity levels of the organizations from four different perspectives (Organizational–Strategic, Organizational–Tactical, Technological–Strategic, and Technological–Tactical).* The CSFs proposed here were measured in two Venezuelan organizations. It was thereby possible to prove that the CSFs proposed are not alien to the existing integration projects and are a guideline for organizations when implementing their integration projects. Measuring the CSFs shows which ones the company still fails to comply with in terms of the acceptability level. It also pinpoints the CSFs with low integration levels that still do not comply and may impact the integration ability of higher levels. In addition to this, they also provide an opportunity to determine the company's integration levels.

INTRODUCTION

The organizations face a major challenge today: *to integrate the business and its Information Systems (IS)*. All companies having to interact not only with its departments or branches, but also with the whole business chain (dealers, suppliers, customers in other countries, as well as managers who need business statistics from any part of the world), means that now organizations must integrate with entities that are related to their business line. However, achieving this integration is not easy task, which *makes it necessary to have a set of Critical Success Factors (CSFs) and a way of measuring them according to the integration levels of IS,* which guide any company when it has to undertake its integration projects. These CSFs are presented from different four points of view which, according to Esteves and Pastor (2001), enrich their definition, as their approach is not just technological, but organizational as well. These are: Organizational–Strategic, Organizational–Tactical, Technological–Strategic and Technological–Tactical.

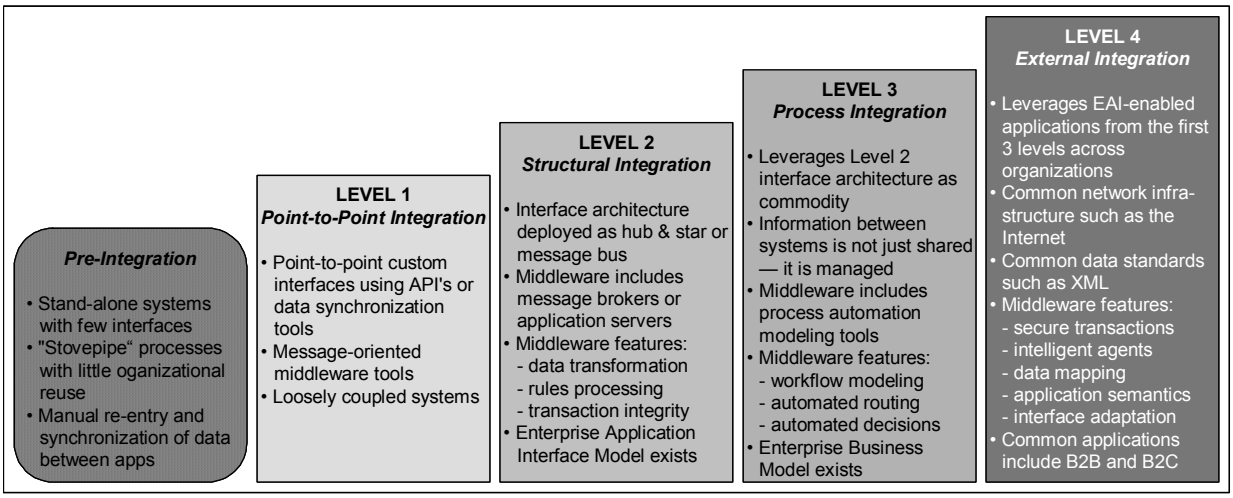
Given the foregoing, in *this research a set of CSFs are proposed, classified and evaluated, according to the maturity levels of the organizations' integration proposed by Schmidt (2000).* These levels are based on the company's existing Information Technologies (IT). It is therefore necessary to see what IS they have and the scope of them, if there is access to data analysis, if the business process flow is optimal, how they inter-relate to the environment, what electronic business technologies they have, among other things. A way to measure their presence in the organization is also proposed. The most important contribution of the proposed CSFs is offering a reference framework, from different perspectives, taking into account IT necessary for the business integration.

This set of CSFs and their measurement were evaluated in two case studies of Venezuelan companies, showing that these factors serve as the basis for determining whether the company meets the desired acceptability levels for each CSF. This article, in addition to this introduction and its conclusions, comprises four sections; first of all describing the maturity levels according to Schmidt (2000), second, proposing the CSFs, third, describing how the CSF were measured, and fourth, analyzing the results.

MATURITY LEVELS

The Maturity Levels proposed by Schmidt (2000) are: *Level One: Point-to-Point Integration, Level Two: Structural Integration, Level Three: Process Integration and Level Four: External Integration.* Figure 1 summarizes these levels and their characteristics.

FIGURE 1
INTEGRATION OF APPLICATIONS. MATURITY MODEL (Schmidt, 2000)



They show clearly what an organization's maturity level is in terms of integration, by relating the levels to the extent of integration to be achieved; in other words, at the Business or Systems level, which is more understandable for an IT manager. It also ranks the organization. The highest levels require an organization with a sound IT infrastructure and need the technologies used for the integration (Enterprise Resource Planning-ERP, Data Warehouse, Customer Relationship Management-CRM, etc.) to be successfully implemented. One can see that at a particular level a company can have Information Systems Integration or Business Integration, but the fact that it may already have Integration Technologies in place cannot be taken to mean it has reached a given level of integration.

Upon analyzing the four integration levels defined by Schmidt (2000) it becomes clear that the higher the level the larger the investment needed, and business strategies are used too (CRM, Supply Chain Management-SCM) as well as technologies that involve making radical changes to the organization's processes. These whole series of aspects is what makes Systems and Business Integration a fairly difficult path to follow, but it is also a necessary one in order to survive in today's changing environment.

PROPOSED CRITICAL SUCCESS FACTORS

A CSF is "*...the limited number of areas in which the results, if satisfactory, will ensure successful behavior for organizations*" (Esteves and Pastor, 2001). Rockart (1979) was the first author to apply the CSFs approach in the field of IS. The purpose was to provide Chief Executive Officers (CEOs) with the necessary information on the organization's critical characteristics and which IS might be developed to meet that need.

In order to define and evaluate the CSFs according to the Levels of Integration proposed in this article, the model developed by authors Esteves and Pastor (2001), which is called the "Unified CSF Model" will be used, since it allows the critical factors to be classified from four perspectives. This model was successfully defined for ERP applications, based on a list of CSFs prior to their investigation. The model unifies the CSFs from four different perspectives (Esteves and Pastor, 2001):

- *Organizational Perspective*: related to everything concerning business processes, structure and organizational culture.
- *Technological Perspective:* focused on aspects related to a particular product; i.e. connected with technical aspects such as hardware or software needs.
- *Strategic Perspective*: related to the organization's mission and long-term goals.
- *Tactical Perspective*: affects business activities and short-term objectives.

The CSFs to be considered according to each level of integration (Smith, 2000), and on the basis of the model formulated by Esteves and Pastor (2001), are proposed below.

CSFs Level 1: Point-to-Point Integration

The main characteristic of this level is: poorly linked systems, personalized interfaces, message-oriented middleware. There is a basic infrastructure for exchanging information between applications, though it lacks any real business intelligence. Table 1 shows the factors defined to attain this level.

TABLE 1
CSF FOR LEVEL 1

	Strategic	Tactical
Organizational	• Effective project leadership • Project management	• Administrative support for the project
Technological	• Technological infrastructure	• Adequate communication software configuration

CSFs Level 2: Structural Integration

The main characteristics are: at the "middleware" level the business rules and data transactions between applications are consolidated, the organization establishes an interface integration model in keeping with a common data model, among others. Table 2 shows the factors defined in order to attain this level.

TABLE 2
CSF FOR LEVEL 2

	Strategic	Tactical
Organizational	• Outsourcing management • Users participation • Effective project leadership • Project management	• Administrative support for the project
Technological	• Technological infrastructure • Internal and external training plan	• Data model documentation, unification and updating.

CSFs Level 3: Process Integration

At this level a common business model is shared by the entire organization. In addition to that, powerful tools are used to implement the business in the middleware layer, supporting easy-to-use procedure modeling capacities, workflow functionalities and decision support tools. Table 3 shows the factors defined in order to attain this level.

TABLE 3
CSF FOR LEVEL 3.

	Strategic	Tactical
Organizational	• Organizational structure • User participation • Effective change Management • Project management • Impact of Information Systems on the organization • Effective project leadership • Implementation strategy	• Administrative support for the project
Technological	• Technical support • Internal and external training plan • Technological infrastructure	• Expert project team

CSFs Level 4: External Integration

At this level, the companies are evaluated real business applications, the influence of technology, the transformation of business processes and the new structures for redefining the organization focusing on the customer. Table 4 shows the factors defined to attain this level.

TABLE 4
CSF FOR LEVEL 4.

	Strategic	Tactical
Organizational	• Effective change management • User participation • Impact of Information Systems on the organization • Determine and justify change at the level of productivity • Senior Management support • Implementation strategy • Administration of the scope of the project. • Effective project leadership • Project management	• Outwards and inwards communication • Administrative support for the project
Technological	• Internal and external training plan • Security • Technical support • Technological infrastructure	• Expert project team

Some of the CSFs proposed can be seen to be repeated at each level and some are repeated at several levels. This is because they do not belong to the integration level, but to the integration process itself. These are shown in Table 5.

TABLE 5
GENERAL CSFs.

Critical Success Factor	Levels	Type
Administrative support for the project	All	Organizational - Tactical
Technological infrastructure	All	Technological - Strategic
Effective project leadership	All	Organizational - Strategic
Project management	All	Organizational - Strategic
User participation	Level 2, Level 3 y Level 4	Organizational - Strategic
Internal and external training plan	Level 2, Level 3 y Level 4	Technological - Strategic
Effective administration of organizational changes	Level 3 y Level 4	Organizational - Strategic
Impact of Information Systems on the organization	Level 3 y Level 4	Organizational - Strategic
Implementation strategy	Level 3 y Level 4	Organizational - Strategic
Expert project team	Level 3 y Level 4	Technological - Tactical
Technical support	Level 3 y Level 4	Technological - Strategic

Once the CSFs have been presented, it is necessary to describe the way they can be measured in a company that decides to undertake integration projects.

EVALUATION METHOD OF THE CSFs

The CSFs were evaluated through the evaluation method suggested by DESMET, which enables methods to be selected to evaluate their use in the field of Software Engineering (Kitchenham et al., 1997). DESMET puts forward nine evaluation methods and a series of criteria to help choose the right method (Kitchenham et al., 1997). The result of this step is the most appropriate evaluation method to apply in the research. In this case, the method with the highest percentage (80%) was: *Features Analysis – Case Study*, this one has 6 steps, which are:

1. *Choosing the method/tool to be evaluated*: for this specific aspect the CSFs were evaluated according to the Integration Levels of the Systems in Venezuelan Organization defined above which are expected to be of use to organizations wishing to conduct a research project in their business. In order to do so, information is required on the level of integration it is at and whether the conditions are right to for moving on to another level.

2. *Identifying a set of features to be evaluated and producing an evaluation format to work in conjunction with potential users:* in this case the features to be evaluated were the variables of the operational definition of each of the CSFs, which indicate whether or not a CSF is present and to what extent. In order to conduct the evaluation of these variables, 4 questionnaires were formulated (measurement instrument chosen), addressed at 4 groups of users: Project Manager, Project Leader(s) or Head(s) of Project, Project Analysts and Users which for this case were chosen with the profile of a Manager.

3. *Selecting one or more pilot projects*: To carry out the evaluation of the CSFs defined, two Venezuelan organizations that are renowned for the quality of the work and their ongoing presence in the domestic market,

which in turn are applications aimed at integrating their IS and their business. Company A markets pharmaceuticals and has been in the market for over 100 years, making it a company with a longstanding tradition; it has several nationwide distributors and is currently developing applications at an *Integration Level 4*; these are: Business-to-Business (B2B) with its Suppliers, B2B with its affiliates, B2C and intranet/extranet. Company B is a leading food and facilities management Services Company, with nationwide presence, but its head office is in Caracas. It is currently an ERP and aiming to integrate its business and in the future move to a higher integration level, based on this new system. This project also involves the company in a change of platform since it will be changing its present software for another totally different one for which its project team has been trained. This application is at the Design stage, positioning the company at an *Integration Level 3*.

4. *Testing the CSFs in the pilot project:* The testing of each of the CSFs was carried out applying the measurement instrument defined in the step 2. These were given to each of the groups to which they were addressed at in each of the companies and the procedure was explained.

5. *Assigning a value to each features of the CSFs*: According to the Features Analysis evaluation, the CSFs defined that will be evaluated were listed, and for each variable of the CSFs a level of conformity was set, and the result showed whether or not the CSF is met in the organization. In order to obtain a value that would enable each CSF to be classified, the percentage of characteristics that the acceptance levels would have to attain was considered, taking into account their importance (Kitchenham et al., 1997).

6. *Analyzing the values obtained and making an evaluation report*: After applying the questionnaire, the results were analyzed and the CSFs were refined.

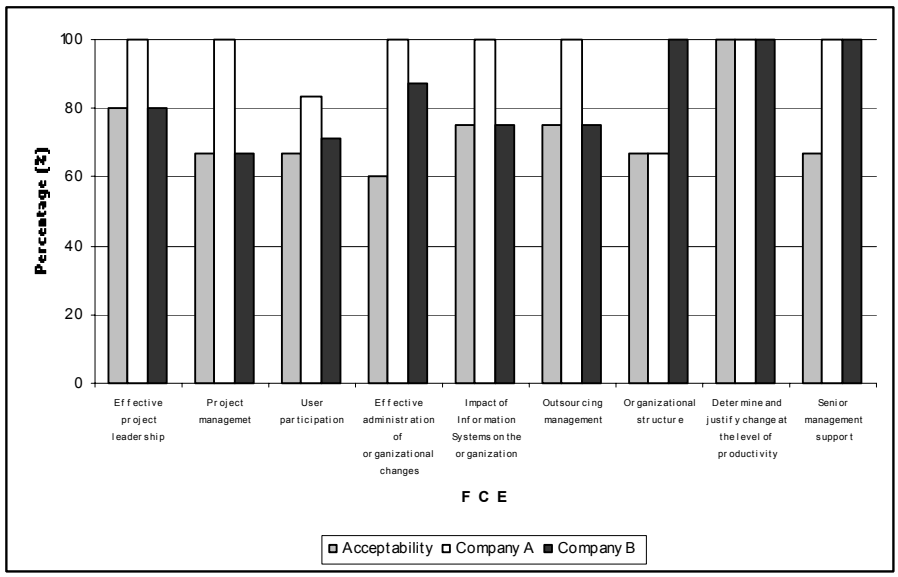
RESULTS OF THE CASE STUDY

The analysis of the different perspectives applied to the companies selected is shown below. Each analysis presents a figure where the minimum acceptability level is compared with the level attained by each company.

1) Organizational–Strategic perspective

Figure 2 shows the level of acceptability by company of all CSFs from the Organizational – Strategic perspective.

FIGURE 2
ACCEPTABILITY VALUE BY CSF OF THE ORGANIZATIONAL–STRATEGIC PERSPECTIVE BY COMPANY

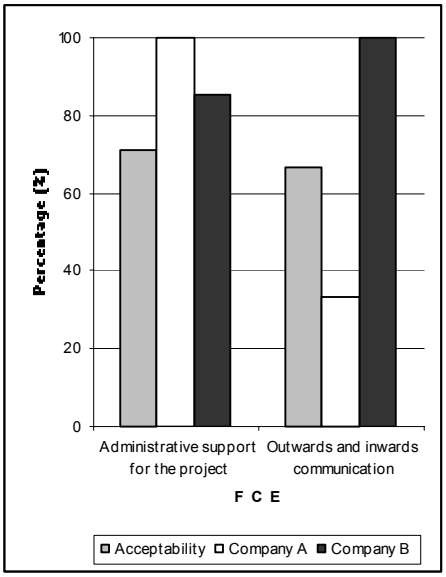


It was seen that *Company A* does not meet the minimum acceptability level (75%) of the *Outsourcing Management* CSF having obtained only 50%. A study of the situation revealed that the measurement variables of the factor related to Documentation were not met; it has either not been updated by the supplier or is non-existent, which shows that at the Organizational–Strategic level the companies studied do not afford sufficient importance to documenting their history with applications or hardware suppliers, or simply to user or procedure manuals, which involves dependence as regards outsourcing, or worse still, the current applications are not documented. Company **A** fulfills the acceptability level of the CSFs from this perspective. However, there are some that are very close to that minimum value, so it is recommended that the projects be reinspected throughout their lifecycle. For example, *Company A* has an acceptability level of 66.66% for the *Organizational Structure* CSF that is equivalent to the minimum accepted. Being Organizational–Strategic this has an immediate impact on the duration of the project, the project management, in other words absence of CSFs at a given point in time would be significant to the project's culmination. The recommendation therefore is to try to lower the decision-making levels.

2) Organizational–Tactical perspective

Figure 3 shows the level of acceptability by company of all CSFs of the Organizational–Tactical perspective.

FIGURE 3
ACCEPTABILITY VALUE BY CSF OF THE ORGANIZATIONAL–TACTICAL PERSPECTIVE BY COMPANY

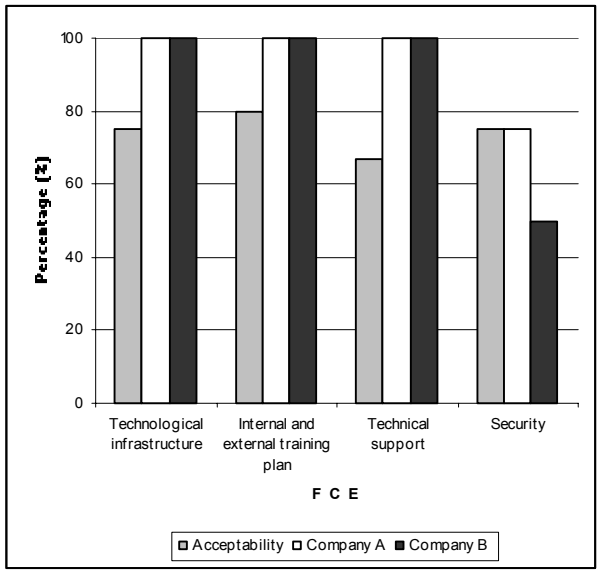


Note that *Company A* does not meet the acceptability level (66.66%) having obtained only 33.33% for the *Outwards communication and inwards communication* CSF, and when the CSF measurement variables that fail to not meet the acceptability level are examined, what can be seen is *Early evidence of electronic communication between the organization and its chain* and *Reliability of security inwards and outwards,* which shows that at this Organizational–Tactical level the organization underestimates the importance or the need for communication tests, above all in level 4 integration projects (CRM, B2B, etc.) where business is integrated through these applications and it is fundamental to pinpoint any communication or integration problem in time, otherwise the project will be unable to fulfill its objectives. As regards security, this is essential as applications such as those mentioned above need to be highly secure in order to protect the organizational data, which must be taken into account right from the start of the project, and not in its latter stages. In the specific case of this company, security must integrate the entire business chain. *Company B* meets the acceptability level of the two CSFs of this perspective.

3) Technological–Strategic perspective

Figure 4 shows the level of acceptability by company of all CSFs of the Technological–Strategic perspective.

FIGURE 4
ACCEPTABILITY VALUE BY CSF OF THE TECHNOLOGICAL-STRATEGIC PERSPECTIVE BY COMPANY



Note that *Company A* meets the acceptability level of CSFs belonging to this perspective. *Company B* does not meet the *Security* CSF, and even *Company A* only meets the minimum expected, which shows that both companies should look at this CSF a little more carefully since, as mentioned in the previous perspective, it is key in any application. For this CSF the Security roles must be assigned from the outset and the development of the applications must be established, taking Security into account.

4) Technological-Tactical perspective

Figure 5 shows the acceptability level by company of all CSFs of the Technological– Tactical perspective.

FIGURE 5
ACCEPTABILITY VALUE BY CSF OF THE TECHNOLOGICAL-TACTICAL PERSPECTIVE BY COMPANY

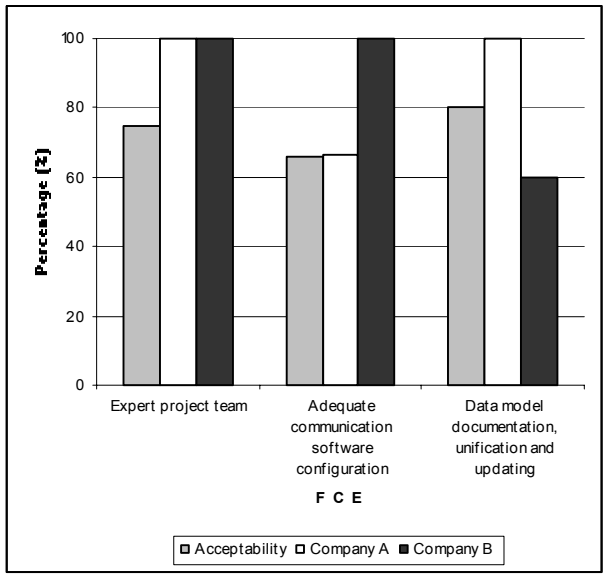


Figure 5 shows that *Company A* meets the acceptability level for each CSF belonging to this perspective and *Company B* does not meet the acceptability level (80%), having only obtained 60% for the *Documentation, Unification and Updating of Data Model* CSF, which is was shown in earlier perspectives. As far as the Documentation in this Company is concerned, since the existing applications were not documented, neither were the data models.

Following the analysis, it was seen from the Organizational aspects, strategic as well as tactical, that the companies selected must improve when a projects ends the development stage and not before. Thus Table 6 summarizes the CSFs that must be improved by the companies chosen, stressing the need to strengthen *Security* in both of them.

TABLE 6
CSF THAT MUST BE IMPROVED BY EACH COMPANY

Critical Success Factor	Company A	Company B
Outsourcing management	No Fulfill	Fulfill
Outwards and inwards communication	No Fulfill	Fulfill
Security	Only fulfills the minimum	No Fulfill
Data model documentation, unification and updating	Fulfill	No Fulfill

It is necessary that from the start of the project the CSFs related to any stage be studied, that the risks involved by estimated and contingency or mitigation plans be put in place.

CONCLUSIONS

Business and Information Systems Integration is now a success factor in today's organizations; however, the complexity and difficulty of the integration projects that must be faced by each company is no secret. An Evaluation of a set of CSFs was proposed according to the integration levels, in conjunction with four important perspectives for the business: Organizational–Strategic, Organizational–Tactical, Technological–Strategic and Technological–Tactical. The CSFs proposed were evaluated through the Features Analysis – Case Study Method in two Venezuelan companies that are currently implementing integration projects. The CSFs are a guide for organizations when they must carry out integration projects, and from the outset they have to consider the factors that are missing or that fail to meet the level of acceptability required. The proposal makes it possible to measure their level of presence in organizations wanting to conduct a research project.

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